

Unit 1 Introduction to AI and AI agents

Q1 Artificial intelligence goals, scope, applications?

- AI is the ability of a computer or a machine to think and act like humans.

Abilities like-

- Problem solving
- Learning
- Decision making
- Understanding languages
- Recognizing images and voice

Goals of AI :-

1. Automation
2. Problem solving and Reasoning
3. Learning from data
4. Understanding Natural language
5. Perception
6. Robotics
7. General intelligence

Application (Branches) of AI :-

1. Machine Learning :-

- Enabling machine to learn data

Ex:- Netflix recommendations

2. Natural Language Processing (NLP) :-

- Help machine to understand and response to human language.

Ex:- Chat GPT, Siri

3. Robotics :-

- Design of intelligent robots

Ex:- Humanoid robot like Sophia.

4. Neural network (Neural) :-

- Understand human brains (neurons) to recognize complex patterns.

5. Deep learning :-

- Multi-layered neural networks for high level learning.

Future and Scope of AI :-

1. Healthcare
2. Education
3. Transportation
4. Business and Finance
5. Robotics and automation
6. Cyber-security
7. Daily life applications

Q2 History (Evolution of AI)?

1. 1940s - Early Foundations?

- First computer were built (ENIAC).

2. 1950s - Alan Turing Test?

- Turing test for testing if a machine can think like a human.

3. 1956 - Birth of AI as a field?

- John McCarthy used the term 'AI'.

4. 1960-70 - Early AI Programme.

- ELIZA (early chatbot)

5. 1980 - Machine learning boom.

- Expert system made and using in business.

6. 1997 - AI Defeats human in chess

- IBM deepblue defeated champion Garry Kasparov.

7. 2000 - Rise of Data and Internet

- Search engines, recommendation systems.

8. 2010 - Deep learning Era?

- Image recognition, NLP, self driving cars.

9. Present and future:-

- AI is now used in healthcare, education, robotics.

Q3 Agent and Intelligent agents?

- An agent is anything that perceives the environment through the sensors and acts on that environment using actuators.

1. Sensors - input device (camera, microphone)

2. Actuators - output device (Speaker, robotic arms).

Ex:- like self driving cars.

Intelligent agent :-

- An intelligent agent is an agent that acts rationally → means it chooses the best possible action to achieve its goals.

Characteristics :-

1. Autonomy - without human intervention, improve with experience.
2. Perception and Environment - It must sense and environment accurately.
3. Rational decision making - It should always choose best possible outcome.
4. Adaptability and learning - It should learn from past action/data.
5. Goal oriented behaviour - It must act to achieve specific goals effectively.

Types of intelligent agent :-

1. Simple reflex agent
2. Model based reflex agent
3. Goal based agent
4. Utility based agent
5. Learning agent

Q4 Types of environment in AI?

1. Fully observable vs Partially observable
2. Deterministic vs Stochastic
3. Episodic vs sequential
4. Static vs Dynamic
5. Discrete vs continuous
6. Competitive vs collaborative
7. Single agent vs multi agent
8. Known vs unknown

Q5 Problem solving approach of AI Problem?

- When AI solves a problem, it needs some imp. components (parts). These are like the building blocks of AI.

AI components required to solve a problem -

1. Learning
2. Knowledge representation
3. Reasoning
4. Problem solving
5. Perception
6. Language understanding (NLP)
7. Planning
8. Execution (Action)

Q6 Problem solving agents and steps?

- A problem solving agent is an AI agent that finds the best sequence of actions to reach a goal. It works step by step.

Steps :-

1. Define the goal
2. Define the problem
3. Search for a solution
4. Choose the best action
5. Execute the action
6. Observe the new state
7. Repeat until the goal is reached.

Ex! Suppose the agent is a taxi driver and goal is 'Drop passenger at the destination'.

Unit 72 Problem solving and Search Strategies

Q1 Uninformed Search Strategies?

- Uninformed search means -
- We do not have extra information about the goal
- We only know the start, goal state and possible moves.
- That's why it is also called blind search.

Ex:-

1. BFS
2. DFS
3. DLS
4. Iterative deepening

Q2 Informed search strategies?

- These are that techniques that uses extra knowledge (heuristics) about the problem to find the solution more efficiently.
- Unlike uninformed which knows nothing except the problem definition.
- It uses heuristic function ($h(n)$) that estimates the cost or distance from the current node to the goal.

Ex:-

1. A* algorithm
2. Heuristic DFS
3. Best First Search

Q3 Heuristic and Heuristic search?

- Heuristic = smart guess / shortcut rule that helps us make decisions faster without checking everything.

Components of heuristic search:-

- Initial state
- Goal state
- Actions
- Path cost $g(n)$
- Heuristic function $h(n)$
- Evaluation function $f(n)$

Ex:-

1. Greedy Best first search
2. A* Search
3. Hill Climbing
4. Simulated Annealing

Q4 Difference b/w Uninformed and informed Search

Uninformed Search	Informed Search
• Search without any extra information • No knowledge about the goal direction • Time consuming • More complexity (Time, Space) • Ex:- DFS, BFS etc	• Search with additional information (heuristic) • Use knowledge / heuristics to find step to solution • Provide quicker solution. • Less complexity (Time and Space) • Ex:- A*, Best first search

Q5 BFS (Breadth First Search):- (Complete)

- BFS explores the search tree level by level
- It start from root node, visit all children first then move to the next level.
- It uses queue (FIFO) data structure.

Properties:-

- will always find a solution if one exists (complete).
- optimal - find always shortest path
- Time complexity: $O(b^d)$ → d - depth.

Algorithm (BFS):-

- 1: Put the start node in a queue.
- 2: Repeat until the queue is empty -
 - Remove the first node
 - If it is the goal → return solution
 - otherwise, expand the node and add all the children to the queue.

Q6 DFS (Depth First Search) (Incomplete)?

- DFS explores as deep as possible along one branch before backtracking.
- It uses a stack (LIFO) data structure.

Properties-

- Not Complete (may go infinite in deep tree)
- Not optimal (may not give shortest path)
- Time complexity: $O(b^m)$
 - b - branching factor
 - m - maximum depth.

Algorithm (DFS):-

- 1: Put the start node in a stack.
- 2: Repeat until the stack is empty -
 - Pop the top node
 - if it is the goal → return solution
 - otherwise expand the node and push children onto the stack.

Q6 Greedy Best First search?

- It is informed search technique.
 - Greedy means it always tries to go towards the goal as quickly as possible.
 - It uses only heuristic function $h(n)$.
 - It doesn't care about the cost so far $g(n)$.
 - GBFS always chooses the path that looks closest to the goal according to the heuristics function.
- Formula used - $f(n) = h(n)$

Working:-

1. Start from the initial Node
2. Evaluate path using heuristic function
3. Choose the best path
4. Repeat the process.

Q7 A* Search algorithm?

- A* is an informed search algorithm.
- It finds the best and shortest path from the start node to goal.
- It uses both -
 1. $g(n)$: cost to reach the current node (best known)
 2. $h(n)$: Estimated cost from current to goal (future guess using heuristic)

Formula used - $f(n) = g(n) + h(n)$

Algorithm:-

- Start from initial node.
- find $f(n) = g(n) + h(n)$ of every neighbour.
- choose that node that has minimum value of $f(n)$.
- Repeat this process until reach the goal.

Q8 Hill climbing? Algorithm?

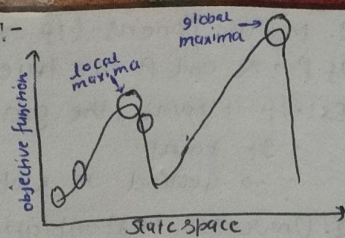
- Hill Climbing is a heuristic search algorithm in AI to find the best solution by step by step improving the current situation.
- It is designed to solve problems where goal is to find an optimal solution by iteratively moving from current state to a better neighboring state according to a heuristic or evaluation function.

Hill climbing algorithm Steps:-

1. Initial State
2. Neighboring State
3. Move to neighbor
4. Termination

Algorithm:-

- Start from initial state
- Look at all neighbors (next possible moves)
- Choose the best one (highest value)
- If no better neighbor exists $\rightarrow$ Stop (you are at peak)
- Otherwise $\rightarrow$ Move to the neighbor and repeat.



Limitations of Hill Climbing:-

1. Local Maxima problem
2. Plateau problem
3. Ridge problem

Q.9 Constraint satisfaction problem (CSP)?

A constraint satisfaction problem is a problem where-

1. you have a set of variables
2. Each variable has a domain (possible value).
3. There are constraint (rule/condition).

goal: Assign values to variables such that all constraints are satisfied.

Ex:-

- Sudoku, crossword, Map coloring

Components:-

1. Variables
2. Domain
3. Constraints (Rules)

Q.10 Alpha-Beta Pruning?

- Alpha-Beta pruning is an improvement of min max algorithm.
- It does not change the final result.
- It skips checking some branches of the game tree that will never affect the final decision.
- This saves time and effort.

Key idea:-

- Alpha (α) $\rightarrow$ best (maximum) value for the MAX player.
- Beta (β) $\rightarrow$ best (minimum) value for the min player.
- if $\alpha \geq \beta \rightarrow$ Pruning (Stop exploring that branch).
- Time complexity $\rightarrow O(b^{d/2})$ (Best case)
 $O(b^d)$ (worst case)

Steps of alpha beta pruning:-

- Step 1: Build the game tree
 - Draw all possible moves.
- Step 2: Evaluate terminal nodes
 - assign utility values to leaf nodes.
- Step 3: Traverse the tree with α and β
 - start from root with $\alpha = -\infty$ and $\beta = +\infty$
 - for MAX nodes
 - update $\alpha = \max(\alpha, \text{value of child})$.
 - if $\alpha \geq \beta \Rightarrow$ Pruning (Prune remaining branches)
 - for MIN nodes
 - update $\beta = \min(\beta, \text{value of child})$
 - if $\alpha \geq \beta \Rightarrow$ Prune remaining branches.
- Step 4: choose optimal Move
 - At the root node, MAX chooses the move with the highest value, but fewer nodes are visited due to pruning.

Unit-73 Knowledge representation and reasoning

Q.1 Propositional logic?

- Propositional logic is the simplest form of logic used in AI.
- It deals with statements (propositions) that can either true (T) or false (F).
- It does not deal with inside the statement only with the whole statement's truth value.

Properties of propositional logic:-

1. Truth Value - either true (1) or false (0)
2. Compositionality - complex statement can built with operator
3. Precedence - NOT > AND > OR > $\rightarrow$ > $\leftrightarrow$
4. Equivalence - Some statement mean the same.

Application of propositional logic-

1. Knowledge representation
2. Automated reasoning
3. Problem solving and planning.
4. Decision making
5. NLP (Natural language processing)

Limitations of Propositional logic-

1. Not expressive enough
2. Scalability problem
3. Binary Only
4. Not quantifiers (like for all $\forall$)
5. No uncertainty handling.

Q.2 First Order logic (FOL)?

- Propositional logic only says whether a whole sentence is true or false.
- FOL goes deeper: it talks about objects, their properties and relationships b/w them.
- That's why FOL is also called Predicate logic.

Key components of FOL:-

1. Constants - Names of object / things
2. Variables - Placeholders for any object (x, y, z)
3. Predicates - Show properties of objects or relations b/w them.
4. Functions - map the object to another object.
5. Quantifiers - how many object -
 - (a) Universal quantifier ($\forall$) $\Rightarrow$ For all.
 - (b) Existential quantifier ($\exists$) $\Rightarrow$ There exists.
6. Logical connectives-
 - AND ($\wedge$)
 - OR ($\vee$)
 - NOT ($\neg$) ($\neg$)
 - implies ($\rightarrow$)
 - iff ($\leftrightarrow$)

Application of FOL:-

1. Knowledge representation
2. Automated Theorem proving
3. NLP
4. Expert systems
5. Semantic web

Syntax of FOL:-

• The syntax in FOL tells us how to write valid statements in FOL using symbols and rules.

1. Terms in FOL (represent objects)-

- constant
- variable-
- functions

Ex: Father (John) means "the father of John"

2. Predicates in FOL (describes properties/relations):-
Predicates = statements about objects-
Ex:- $\text{Human}(x) \rightarrow "x \text{ is a human}"$.
3. Quantifiers in FOL - (talk about many objects)
 - (i) Universal ($\forall$): For all
 - (ii) Existential ($\exists$): There exists at least one.
4. Connectives in FOL (combine statements):-
• $\neg$ • $\vee$ • $\rightarrow$ • $\leftrightarrow$ • $\neg$
5. Well formed formulas (WFFs):-
• valid statements in FOL made by combining terms, predicate, quantifiers and connectives.

Semantics of FOL:-

- It means How to give meaning to formulas (deciding whether a statement is true or false in a given situation or model)
- It explains what the symbols stands for and when a formula is true.

1. Key elements of Semantics:-

- Domain of discourse
- Variables
- Constants
- Functions
- Quantifiers ($\forall, \exists$)
- Predicates

2. Interpretation (Model):-

- means how to map symbols to real meaning.

Domain

- Interpretation of constants
- Interpretation of predicates
- Interpretation of functions

3. Truth in FOL:-

- A formula is true or false depending on the interpretation.

4. Quantifiers:-

- Universal ($\forall$)
- Existential ($\exists$)

5. Satisfaction:-

- A formula is satisfied if it is true under a given interpretation.

6. Validity:-

- A formula is valid if it is true in every interpretation.

7. Relation b/w Satisfaction and Validity:-

- (i) Satisfaction - true in some model
- (ii) Validity - true in all models.

0.3 Inference and Inference rules?

- In FOL inference is the process of deriving new true statements from existing ones using logical rules.

Ex:- Fact: $\forall x (\text{human}(x) \rightarrow \text{Mortal}(x))$

• Fact: $\text{Human}(\text{Socrates})$.

• Inference: $\text{Mortal}(\text{Socrates})$.

Inference Rule in FOL:-

- These are formal rules used by AI to reason:

1. Modus Ponens (If-Then Rule):-

If $P \rightarrow Q$ and P are true $\rightarrow$ then Q must be true.

Ex:- If it rains, the ground gets wet.

• It rains

• $\rightarrow$ Ground is wet.

2. Universal Instantiation (UI):-

• From $\forall x P(x)$, we can take any object 'a' and say $P(a)$.

Ex:- $\forall x \text{ Human}(x) \rightarrow \text{Mortal}(x)$

Take Socrates: $\text{Human}(\text{Socrates}) \rightarrow \text{Mortal}(\text{Socrates})$

3. Existential Instantiation (EI):-

From $\exists x P(x)$, we can say $P(c)$ for some new constant c .

Ex:- $\exists x \text{ loves}(x, \text{ice cream}) \rightarrow \text{Loves}(c, \text{Icecream})$.

4. Resolution Rule:-

• used in automated theorem proving $\rightarrow$ combines clauses to remove contradictions.

Ex:- $P \vee Q$

• $\neg P$

• $\rightarrow Q$

Q. Explain Wumpus World Problem?

• Wumpus world is a grid based game environment.

• The agent is in (4x4) grid room where some rooms are safe while other are dangerous.

• The goal of agent-

• Find the gold

• Avoid danger

• Safely come back out of the rooms.

4.1	4.2	4.3	4.4
S		B	Pit
3.1	3.2	3.3	3.4
Agent	S, B gold	Pit	B
2.1	2.2	2.3	2.4
S		B	
1.1	1.2	1.3	1.4
Agent	B	Pit	B

Elements in Wumpus world:-

1. Wumpus
2. Pits
3. Gold
4. $S \rightarrow$ Stench
5. $B \rightarrow$ Breeze
6. Agent
5. Walls

PEAS description of Wumpus World:-

• It describes how an agent interacts with the Wumpus world.

1. Performance measures -

- +1000 $\rightarrow$ Agent get the gold and return safely
 - -1000 $\rightarrow$ Agent dies
 - -1 $\rightarrow$ For every move
 - -10 $\rightarrow$ if agent use arrow
- goal: maximize total score.

2. Environment:-

• A cave with 4x4 grid = 16 rooms

Hazards -

- Wumpus (dangerous monster)
- Pits (Fall $\rightarrow$ death)

Clue -

- Stench near Wumpus
- Breeze near pits
- Glitter in gold room

3. Actuators :-

- Move Forward
- Turn Right / Left
- Shoot Arrow
- Grab gold
- Release

4. Sensors -

- Breeze → Pit nearby
- Stench → Wumpus nearby
- Glitter → gold in current room
- Scream → Wumpus has been killed.

How agent operates with PEAS :-

1. Perception
2. Inference
3. Planning
4. Action
5. Repeat until success

Q.5 Forward vs backward chaining?

Feature	Forward	Backward
• Driven by	• Data (fact)	• Goal
• Start from	• Facts	• Goal
• Best for	• Situations with lots of data to explore all possible outcomes	• Specific problems where goal is already known
• Ex: use	• Medical diagnosis (finding all possible diseases)	• Troubleshooting (finding why some thing failed)
• Forward chaining → Start with fact → Move forward		
• Backward chaining → Start with goal → Move backward		

Q.6 Ontologies in AI and Ontologies Engineering?

- Ontology is like a map of knowledge that tells us -
- What kind of things exists in a domain (movies, actors, scripts etc).
- What properties they have (runtime, age, rating)
- What they are related (actor act in a movie, director direct a movie).
- So ontology means creating a structured map of knowledge where we identify things, their groups, their properties and their relationships.
- Ontology = categories + properties + relationships that helps computers understand meaning not just words.

Why it is needed :-

- To give computers a clear understanding of knowledge.
- To make information organized, reusable and sharable.
- To help in AI, expert system, knowledge based application.

Step in ontological Engineering -

1. Decide the domain
2. List important concepts.
3. Define categories
4. Define properties

3. Define relationships

- Test and Refine the Ontology.

Ontological Reasoning -

- Ontological reasoning means ontology to make logical conclusions.
- It is the process where the systems reasons (thinks) using the structured knowledge in ontology.

How it works -

- The computer checks the relationships and rules in the ontology.
- Then it derives new facts or answer queries.

Ex:-

- Knowledge is Ontology
- Amir Khan is an actor
- An actor is a type of person
- Reasoning result - Amir Khan is also a person.

Unit -4 Uncertainty and Learning Techniques

Q.1 Uncertainty and Sources of uncertainty?

- Uncertainty means not being 100% sure about something.
- Uncertainty in AI means lack of complete confidence in decisions.
- Uncertainty comes because -
 - Data may be incomplete or wrong
 - Models may not be perfect
 - Real world is complex and unpredictable

Sources of uncertainty in AI :-

1. Data uncertainty
 2. Model uncertainty
 3. Algorithmic un.
 4. Environmental
 5. Human
 6. Ethical
 7. Legal
 8. Uncertainty in AI reasoning
 9. Uncertainty in AI perception
 10. Uncertainty in AI communication
- Ex: chatbot misinterpret user's questions.

Q.2 Probabilistic Reasoning?

- Probabilistic reasoning means making decisions or predictions when we are not 100% sure but have some probability (chances).
- Instead of saying Yes/No it say how likely something will happen.
- So probabilistic reasoning help the AI handle uncertainty.

Key concept in probabilistic reasoning -

1. Probability
2. Random Variables
3. Prior probability
4. Conditional probability
5. Bayes rule
6. Joint probability
7. Inference.

Why do we need probabilistic reasoning -

- Real world is uncertain
- Deterministic (yes/no) reasoning is not enough
- Help AI make better prediction
- More realistic decisions - like human.

Applications :-

- Medical diagnosis
- Self driving cars
- Game and sports
- Weather forecasting
- Spam filtering in email
- Image / Speech recognition

Q.3 Explain Bayes theorem (Rule) with example?

Bayes theorem is a mathematical formula used to update the probability of a hypothesis based on a new evidence. It provides the structured way to revise prior belief when presented with additional data.

Conditional probability -

It is the probability of one event happening given that another event has already happened.

Like - It tells 'what is the chance of A if we already know B has happened'.

Bayes Theorem formula -

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)}$$

Where -

1. $P(A/B) \Rightarrow$ Posterior probability

$\Rightarrow$ The probability of A being true, given B is true.

2. $P(B/A) \Rightarrow$ Likelihood

The probability of B being true, given A is true.

3. $P(A) \Rightarrow$ Prior probability

The probability of A being True.

This is the knowledge.

4. $P(B) \Rightarrow$ Evidence

The probability B being true.

Marginal probability of evidence.

Ex:-

Problem - An animal shelter -

- 60% of animals are dogs
- 40% of animals are cats

It is known that -

- 70% of dogs bark
- 20% of cats bark

An animal is chosen at random and it is heard barking.

$\Rightarrow$ What is the probability that the animal is a dog.

Step 1: Define events -

- D = Animal is a dog
- C = Animal is a cat
- B = Animal barks

Step 2: Given probabilities -

- $P(D) = 0.6$
- $P(C) = 0.4$
- $P(B/D) = 0.7$
- $P(B/C) = 0.2$

Step 3: Find $P(B)$

$$P(B) = P(B/D) \cdot P(D) + P(B/C) \cdot P(C)$$

$$P(B) = (0.7 \times 0.6) + (0.2 \times 0.4)$$

$$P(B) = 0.42 + 0.08 = 0.50$$

Step 4: Apply Bayes theorem -

$$P(D/B) = \frac{P(B/D) \cdot P(D)}{P(B)}$$

$$P(D/B) = \frac{0.7 \times 0.6}{0.5}$$

$$P(D/B) = 0.84 = 84\%$$

Conclusion:-

If an animal is barking, there is an 84% chance it is a dog.

Q.4 Machine learning and types of machine learning?

ML is a branch of AI that allows computers to learn from data without being explicitly programmed.

Instead of giving fixed instructions, we give the system data and let it find patterns and make predictions or decisions.

Key components of ML:-

Data • features • Model • training • testing

Applications of ML:-

Healthcare • Finance • E-commerce • Self-driving cars.

Types of machine learning in AI:-

(i) Supervised - Learn from labeled data

(ii) Unsupervised - learn from unlabeled data

(iii) Reinforcement - Learn by trial and error

Q.5 Supervised Learning vs Unsupervised Learning:-

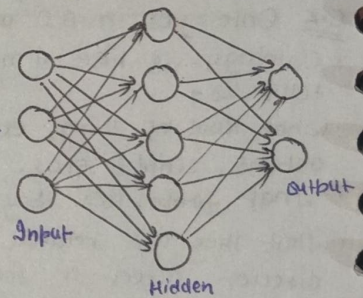
Criteria	Supervised	Unsupervised
Data Labeling	Labeled data	Unlabeled data
Goal	Prediction	Pattern discovery
Complexity	Generally simpler	Can be more complex
Ex.	Classification, regression	Clustering, Association
Real World Use	Spam filter, Siri	Google news, customer segmentation
Algorithm	Decision tree	K-means clustering

Q.6 Artificial neural network (ANN), architecture?

ANN are computer systems designed to mimic how the human brain processes information. Just like the brain uses neurons to process data and make decisions, ANN uses artificial neurons to analyze data, identify patterns and make predictions.

Architecture:-

1. Input layer - This is where the network receives information.
2. Hidden layers - These layers process the data received from the input layer.
3. Output layer - This is where the final decision or prediction is made.



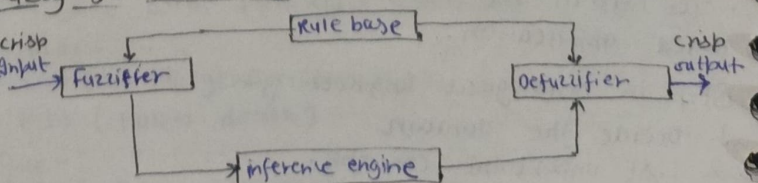
Application of ANN:-

1. Social media
2. Healthcare
3. Finance
4. Personal assistance
5. Marketing
6. Retail

Q.7 Fuzzy logic in ai and importance?

- Fuzzy logic helps work with situations where the information is unclear or partially true.
- Handles uncertainty and vague information.
- Works with partial truth values b/w 0 and 1.
- Helps systems take flexible and human-like decisions.

Fuzzy logic architecture:-



1. Fuzzification:- It is the process of converting precise input values into fuzzy sets.

2. Rule base- A collection of if then rules created by experts to guide the systems decisions.

3. Inference engine- It evaluates the fuzzified input data and applies the relevant rules from the rule base.

4. Defuzzification- Once the inference engine processes the rules and generates the output, it converts these fuzzy result back into precise, crisp value.

Application of fuzzy logic:-

1. Aerospace
2. Automotive
3. Industrial
4. AI
5. Control systems.

Unit 7.5 Applications of AI and Multi agent systems

Q.1 Natural Language processing?

• Natural language processing (NLP) is a branch of AI that helps computers understand, interpret and respond to human language (like Hindi, eng.)

Ex:- Google translation • ChatGPT, Alex, Siri

How NLP Works:-

1. Text input and Data collection
2. Text preprocessing
3. Text representation
4. Feature selection
5. Model selection and training
6. Model deployment and inference
7. Evaluation and optimization
8. Iteration and improvements.

Q.2 What is information retrieval and extraction?

• Information retrieval (IR) is the process of finding useful information from a large collection of data (such as search engines, digital libraries)

Ex: Search on Google web some result shows.

Components of information retrieval:-

• User side (Search process)

1. Problem identification
2. Representation
3. Query
4. Feedback

• System side (Retrieval Process)

1. Acquisition (collecting data)
2. Representation (indexing)
3. File Organization
4. Matching
5. Retrieved Object (Results)

• So IR is like a librarian - user asks book on ML Librarian (System) searches shelves (database) and find matching book, give the best one first.

• Information Extraction:-

• IE is the process of automatically pulling out useful and structured information from unstructured text (like news articles, research paper etc).

Key components of IE:-

1. Text input and preprocessing
2. Named entity Recognition
3. Relation extraction
4. Event extraction
5. Coreference extraction
6. Template filling (Structured output)

Q.3 Robotics and perception planning and motion in Robotics?

• Robotics is the branch of science and engineering that deals with designing, building and using robots.
• A robot is a machine that can sense environment process information and act to achieve tasks (like moving, picking objects, delivering etc).

Perception:-

• Perception means how a robot sees, hears and understand the world around it using sensors. Like, robot sense the environment with, camera, microphones, GPS, touch sensors. and processes with AI + Computer vision + ML to understand the surroundings.

• How robotic perception works (steps):-

1. Sensing:- collect data from environment (camera, touch)
2. Preprocessing:- clean the raw data
3. Feature extraction:- important details picked.
4. Interpretation:- (understanding through AI and ML)
5. Decision making:- take decision on perception
6. Action:- perform the action (execution).

Planning and Motion:-

• Planning = Deciding what path or sequence of actions a robot should take to reach its goal.
• Motion = Actually moving the robot along that path safely and smoothly.
• Together, Planning + Motion help a robot go from start point → goal point without crashing or wasting energy.

How Planning and motion works-

1. Goal definition
2. Perception and mapping
3. Path planning (High level planning)
4. Motion planning (Low level planning)
5. Collision advance
6. Control and execution

In short -

Planning - deciding where to go (like Google map route)
Motion - Actually moving along with that route. together they make robot intelligent and safe mover.

Q.4 Speech recognition?

• Speech recognition is the technology that allows a computer or machine to listen to human speech understand the word and convert them into text or actions.

Ex:- you say hello Siri set the alarm for 7 AM, It understand and creates the alarm.

How it works-

1. Speech input (voice capture)
2. Preprocessing (cleaning the voice)
3. Feature Extraction
4. Acoustic model
5. Language model
6. Decoder
7. Output

Applications of Speech recognition-

- Virtual Assistants (Alex, Siri, Google assistant)
- Transcription tools (converting lectures)
- Voice Commands (car navigation) (Smart home device)
- Accessibility (helping people who can not type)

Components:-

- Microphone
- Preprocessor
- Feature extractor
- Acoustic model
- Lang. model
- Decoder
- Output

Q.5 Software agents and architecture?

- A software agent is like a computer program that works for you automatically.
- It can sense the environment, make decisions and take actions to achieve a goal without you controlling every step.

Ex:- Email spam filter, Chatbots, Recommender systems.

Architecture of software agents:-

It explains-

- How agent perceives the environment.
- How it decides what to do.
- How it acts in the environment.

main component of Agent architecture-

1. Sensors (input)
2. Actuators (output)
3. Agent program (Brain)

Communication:- How agents talk to each other.

- Agents communicate each other to-
- Solve a problem; they need to share information.
- This information sharing called communication.

How do they communicate-

1. Message passing
2. Common language / Protocol
3. Negotiation

Q.6 What is Trust, Negotiation and Bargaining in agents? and Reputation?

1. Trust:- (Ensuring agents are reliable)

- In multi agent systems, not all agents can be trusted blindly.
- Trust means making sure that the information or actions of an agent are correct, safe and not misleading.

Why trust is important:-

1. Correctness
2. Security
3. Reputation systems

2. Multiagent Negotiation:-

- Imagine you have many intelligent agents (like small AI programs or robots) sometimes they need the same resources (money, time, energy, product) or have different goals.
- To solve conflicts - agent negotiate → They discuss, bargain, agree on a solution that works for everyone.

Why do agents negotiate-

- To share limited resources
- To coordinate tasks
- To reach an agreement without wasting resources.

Types of multiagent Negotiation-

1. Cooperative negotiation
2. Competitive negotiation
3. Bargaining
4. Auction based negotiation
5. Contract net protocol (CNP)

3. Reputation in multiagent system:-

- Reputation means the record of past behaviour of an agent.

It is how trustworthy and reliable to decide whether to trust or not.

Ex:- An Amazon rating system.

Why is reputation important-

- 1. Builds trust
- Avoids cheating
- Improve cooperation

How is reputation calculated-

- Agent observe behaviour or receive feedback from others.
- Reputation = Past performance + rating + recommendations
- Good reputation - More trusted
- Bad reputation - less trusted, fewer chances to get a task.

Q.7 What is Explainable AI (XAI)?

• The problem of today AI (Black box) → in normal AI (top part of diagram)-

1. Training data
2. Machine learning process
3. Learned function (Model)
4. Decision / Recommendations

But the problem is: user's don't know how the AI made the decision.

Ex: AI said user is denied a loan.

The user asks why? No clear answer (Black box)

Explainable AI (XAI):-

In XAI (top part of diagram)-

- Training data
- New machine learning process
- Explainable Model
- Explanation interface

• Now the user not only get decisions but also the reason behind it

Ex: AI said loan denied

Then add it! Because income is too low.

Why XAI important:-

- Transparency
 - Trust
 - Debugging
 - Legal and Ethical
- In short:-

Normal AI = Answer only

XAI = Answer + reason

Q.8 Black box and techniques for explaining black box models?

- In black box model we only see the input data, the output data but the process in b/w is hidden or too complex.

Techniques:-

1. LIME (Local interpretable Model-agnostic Explanations):-

- Works by approximating the black box model locally (for one prediction).

2. SHAP (Shapley additive explanations)

- Based on game theory.
- It calculate how much each feature contribute to decision

3. Partial Dependence Plots (PDPs):-

- Shows how changing one decision affect predictions.

4. Feature Importance:-

- Rank features by how important they are for model prediction.

5. Counterfactual Explanations:-

- Answer the ques: (What should be different for diff. outcome)

6. Surrogate Models:-

- Train a simpler model to mimic the black box model's behaviour. easier to understand.